Practical $\LaTeX$

For Academic and Scientific Documents

A practical guide to writing books, theses, question papers, and research articles

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About This Book

Purpose. This book is a practical guide to building academic and scientific documents in LaTeX. It is not an encyclopedia. It explains the project decisions that make books, theses, question papers, research articles, and Hindi manuscripts easier to write, revise, compile, and reuse.

How to read it. Start with the first two chapters if you are new to LaTeX project structure. Then jump directly to the document type you need: book, thesis, question paper, article, tables and figures, or build repair.

Conventions. Commands appear as `\input` or `\includegraphics`. Package names appear as `geometry`, `fontspec`, or `tabularx`. Generated files belong in `gar/`. Figures use labels such as `fig:geometry-visual`; equations use `eq:...`; tables use `tab:...`.

Principle

The best LaTeX file is not the cleverest file. It is the file that can be opened six months later, understood quickly, compiled reliably, and reused safely.

Chapter	Main use
1	Project shape, packages, fonts, page geometry, and Hindi manuscripts.
2	Long documents: books and theses.
3	Shorter academic documents: question papers and research articles.
4	Tables, figures, labels, and references.
5	Builds, reuse, and error repair.

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CHAPTER 1

Project and Preamble

A serious LaTeX document is a small project. The main file controls the document class, packages, page design, fonts, hyperlinks, and the order of included files. Chapter files should carry the writing. Generated files should stay out of the manuscript folders.

Read LaTeX Syntax First

LaTeX source is plain text mixed with commands. A command usually starts with a backslash, required information goes in braces, optional settings go in square brackets, and longer structures use a begin–end pair.

```
\command[optional settings]{required argument}
\begin{environment}[optional settings]
  Content goes here.
\end{environment}
```

Syntax	Meaning
`\section{Title}`	A command begins with `\`. Here the required title is inside braces.
`[10pt,openany]`	Square brackets hold optional settings. If omitted, LaTeX uses defaults.
`{book}`	Braces hold required arguments. The command cannot do its job without them.
`\begin{figure}...\end{figure}`	An environment starts and ends a block such as a figure, table, equation, or list.
`% comment`	A percent sign starts a comment. LaTeX ignores the rest of that source line.
`#1`	Inside a command definition, this means the first argument supplied by the user.

```
\documentclass[10pt,openany]{book}
\usepackage[margin=0.75in]{geometry}

\newcommand{\term}[1]{\textbf{\emph{#1}}}

\begin{document}
\chapter{Introduction}
A \term{macro} is a command that expands to stored instructions.
\end{document}
```

Read the example from left to right: `documentclass` chooses the document type, `usepackage` loads extra features, `newcommand` defines a shortcut, and `begin`/`end` mark the document body. This same grammar is used in articles, theses, books, and presentations.

Compile `main.tex` First

Compile the main file, not an individual chapter file. The main file knows the document class, packages, paths, index setup, and output folder.

Engine	Use it when
`pdflatex`	The document uses standard pdfLaTeX packages such as `fontenc`, `inputenc`, and `lmodern`. It is reliable for most English academic documents.
`xelatex`	The document needs system fonts through `fontspec`, Unicode text, or scripts such as Hindi/Devanagari.
`lualatex`	The document needs system fonts like XeLaTeX, with Lua-based extensions or advanced font control.

On macOS or Linux, run these commands from the project directory:

```
mkdir -p gar
pdflatex -interaction=nonstopmode -halt-on-error \
  -output-directory=gar main.tex
makeindex gar/main.idx -o gar/main.ind -t gar/main.ilg
pdflatex -interaction=nonstopmode -halt-on-error \
  -output-directory=gar main.tex
cp gar/main.pdf main.pdf
```

On Windows Command Prompt, use backslashes for files inside `gar`:

```
if not exist gar mkdir gar
pdflatex -interaction=nonstopmode -halt-on-error ^
  -output-directory=gar main.tex
makeindex gar\main.idx -o gar\main.ind -t gar\main.ilg
pdflatex -interaction=nonstopmode -halt-on-error ^
  -output-directory=gar main.tex
copy gar\main.pdf main.pdf
```

If the source uses `fontspec`, replace `pdflatex` with `xelatex` or `lualatex`. The option `-output-directory=gar` keeps auxiliary files such as `.aux`, `.toc`, `.idx`, and `.log` inside `gar/`.

Part	Why it is written this way
`-interaction=nonstopmode`	LaTeX keeps running instead of stopping at the terminal for every question. This is useful in scripts because the log receives the diagnostics.
`-halt-on-error`	The build stops at the first fatal error. This prevents a broken PDF from looking like a successful build.
`-output-directory=gar`	Auxiliary files and the temporary PDF are written in `gar/`, so the project root stays readable.
`makeindex gar/main.idx`	The index maker reads the raw index entries produced by LaTeX.
`-o gar/main.ind`	The formatted index is written where `\printindex` can read it on the next LaTeX run.
`-t gar/main.ilg`	The index log is stored beside the other generated files.

The two error-handling options are usually used together. The first keeps LaTeX from waiting for keyboard input after an error, which is important in a scripted build. The second stops the build at a fatal error, so a broken run does not leave a misleading partial PDF. Together they make the build automatic but strict: write diagnostics to the log, stop on a real failure, and let the author fix the first cause.

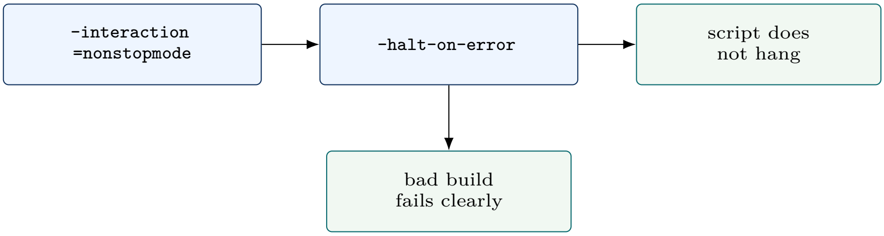


Figure 1.1: The build options make compilation suitable for terminal scripts: no interactive pause, but a clear stop on fatal errors.

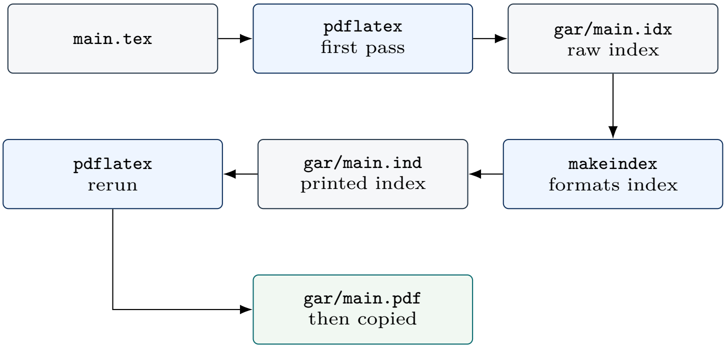


Figure 1.2: The build sequence first writes auxiliary data, then formats the index, then reruns LaTeX so the final PDF can include it.

A Project You Can Reopen

Figure 1.3 is the basic habit. The file `main.tex` should read like the source table of contents. It loads the preamble, starts the document, and includes the major files in order. The writing folders then stay small enough to browse.

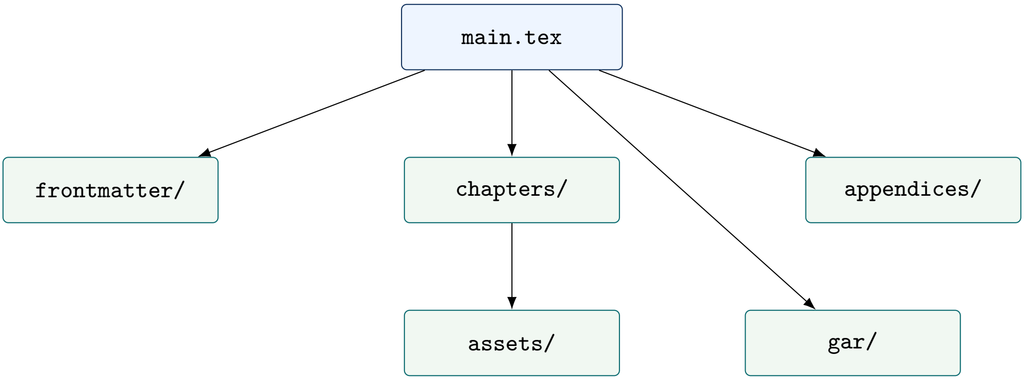


Figure 1.3: A readable project separates control, writing, assets, and generated output.

Location	Role
`main.tex`	Class, packages, page layout, document order, index setup, and metadata.
`frontmatter/`	Title, preface, acknowledgements, notation, or a compact opening note.
`chapters/`	Main explanations, one coherent topic per file.
`appendices/`	Reference material that supports the book but should not interrupt the chapters.
`gar/`	Auxiliary files and build output such as `.aux`, `.toc`, `.idx`, and `.log`.

Commands Used Throughout

The book should not re-explain the same command every time it appears. Use this compact command map once, then apply it in context.

Command	Use
`\documentclass`	Selects the broad document type, such as book, article, report, or thesis class.
`\usepackage`	Loads extra capabilities before the document begins.
`\input`	Inserts another source file exactly where the command appears.
`\chapter` and `\section`	Build the visible hierarchy of the document.
`\includegraphics`	Places an image, normally inside a figure environment.
`\label` and `\ref`	Connect numbered items to cross-references that update after recompilation.
`\index`	Marks a word or topic for the generated index.

Packages by Job

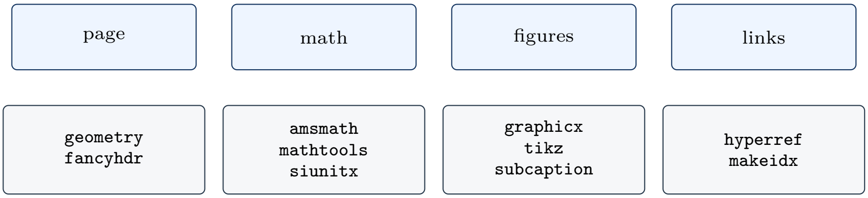


Figure 1.4: Package choices are clearer when each package has a visible job.

`geometry` sets page size and margins. `fontspec` selects system fonts when using XeLaTeX or LuaLaTeX. `amsmath` and `mathtools` improve equations. `siunitx` gives consistent scientific units. `graphicx` inserts images. `subcaption` groups related images. `booktabs` and `tabularx` make readable tables. `tikz` draws diagrams. `hyperref` creates links and PDF metadata.

Page and Font Design

```
\usepackage[
  paperwidth=6in,
  paperheight=9in,
  inner=0.65in,
  outer=0.50in,
  top=0.58in,
  bottom=0.68in
]{geometry}
```

For English documents compiled with pdfLaTeX, `lmodern` is usually enough. For a manuscript that must use a system font, use XeLaTeX or LuaLaTeX with `fontspec`:

```
\usepackage{fontspec}
\IfFontExistsTF{Times New Roman}
  {\setmainfont{Times New Roman}}
  {\setmainfont{TeX Gyre Termes}}
```

Hindi Manuscripts

A Hindi manuscript needs a Unicode engine and a Devanagari-capable OpenType font. The source in `print-hindi/manuscript/book.tex` uses the right pattern: `book`, A5 paper, two-sided margins, `fontspec`, Devanagari font selection, Hindi labels, designed chapter heads, and modular frontmatter/chapter/backmatter files.

```
\setmainfont{Kohinoor Devanagari}[Script=Devanagari]
\setsansfont{Kohinoor Devanagari}[Script=Devanagari]
\linespread{1.16}
\setlength{\parindent}{1.15em}
```

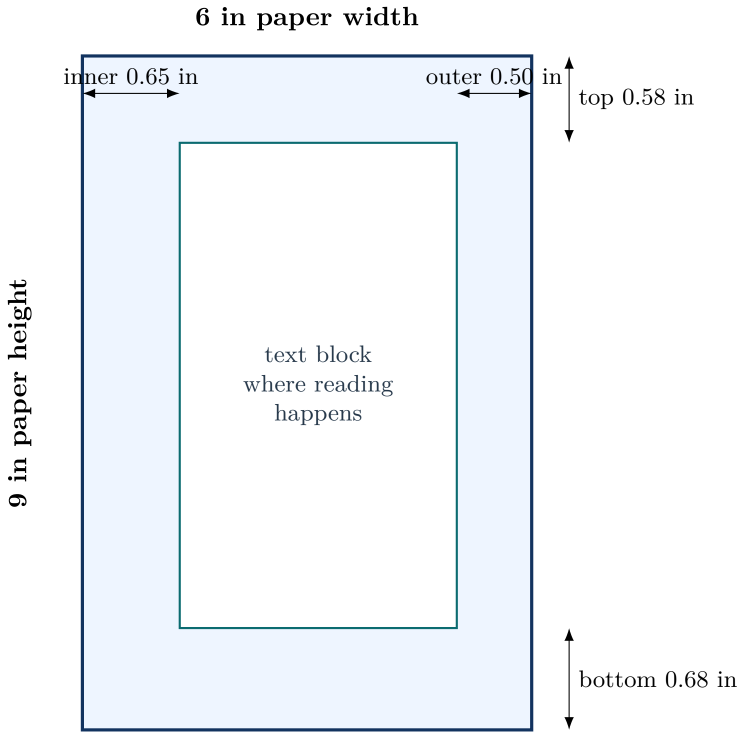


Figure 1.5: The paper is the outer rectangle; the readable text block is inside the margins.

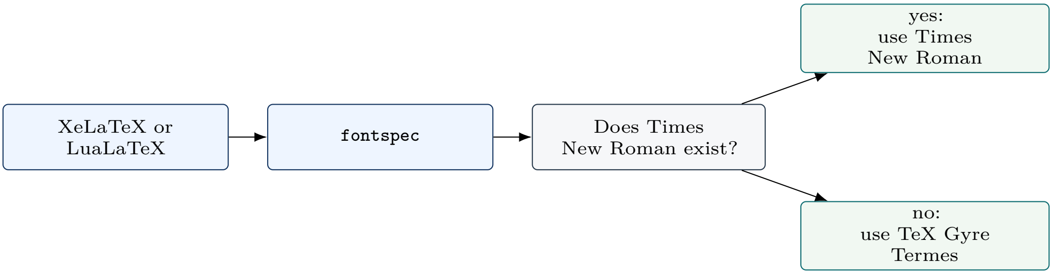


Figure 1.6: The font fallback keeps the source portable: it uses Times New Roman when available and a TeX-compatible Times-like font otherwise.

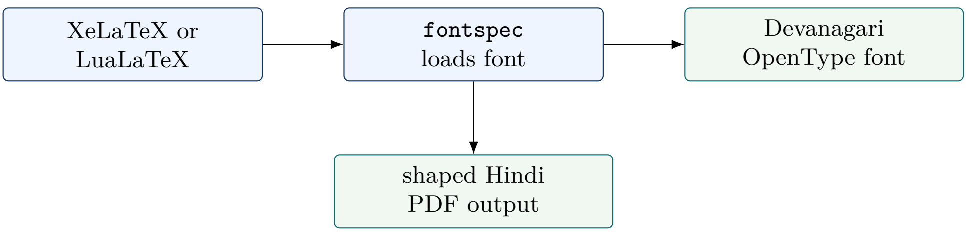


Figure 1.7: Hindi changes the engine and font setup, but not the modular project discipline.

CHAPTER 2

Books and Theses

Books and theses share the same engineering problem: they are long, formal, and revised many times. Their source should make the document order obvious and should keep formal pages separate from the main argument.

The Long-Document Flow

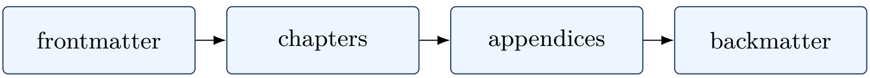


Figure 2.1: Books and theses are easiest to maintain when the source follows the reader's path.

For a book, frontmatter usually means title pages, copyright, preface, and contents. For a thesis, it may also include declarations, certificates, acknowledgements, abstracts, lists of figures, and lists of tables. These formal pages should not be mixed into chapter files.

One Source Pattern

```
\frontmatter
\input{frontmatter/title}
\input{frontmatter/preface}
\tableofcontents

\mainmatter
\chapter{Introduction}
\input{chapters/introduction}

\appendix
\input{appendices/supporting-material}

\backmatter
\printbibliography
\printindex
```

This pattern serves both books and theses. The difference is in the required formal pages, bibliography style, line spacing, fonts, and institutional rules.

Choosing the Document Class

The document class is the first structural decision. It controls the default page style, title behavior, sectioning commands, and whether commands such as `\chapter` exist.

Class	Good use
`book`	Textbooks, monographs, long manuals, and documents with chapters, frontmatter, appendices, and backmatter.
`report`	Theses, project reports, lab reports, and technical documents that need chapters but less book-style frontmatter.
`article`	Papers, question papers, assignments, notes, and short documents without chapters.
`memoir`	Highly customized books where one class should control chapters, headers, captions, and layout.

Do not choose a class by habit. Choose it by the structure the reader will see. If the document needs chapters, use `book` or `report`. If it only needs sections, use `article`.

Book Decisions

A book needs a strong table of contents, consistent chapter openings, useful running heads, and often an index. Packages such as `fancyhdr`, `titlesec`, and `makeidx` support these jobs. Use them to clarify navigation, not to overdecorate the page.

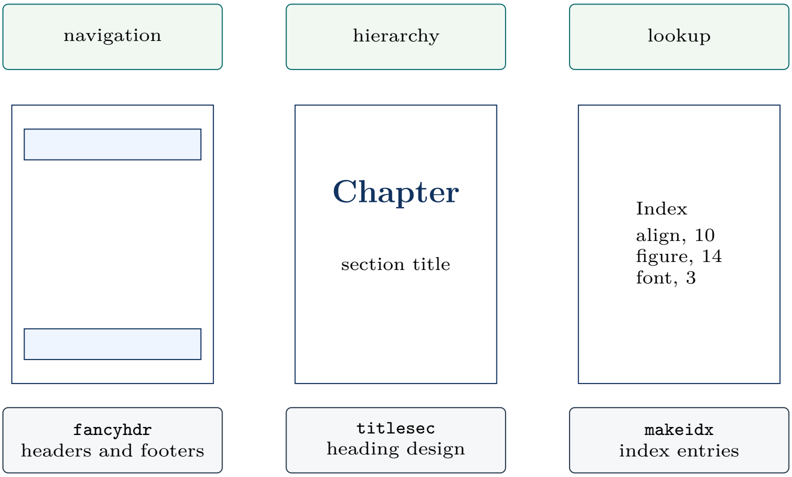


Figure 2.2: Three book-design packages solve three different reading problems: page navigation, heading hierarchy, and topic lookup.

Package	Typical source pattern
`fancyhdr`	Define what appears in the left, right, and center of page headers or footers.
`titlesec`	Redesign chapter and section headings without rewriting the chapter text.
`makeidx`	Activate indexing with `\makeindex`, mark terms with `\index {...}`, then print with `\printindex`.

An index is especially useful in a LaTeX manual because readers often return to a command or package name. Mark only useful terms. An index full of obvious words is harder to use than no index.

Chapter Files in a Book

One chapter file should contain one chapter's prose, figures, tables, and local labels. Keep global design outside chapter files. A chapter should not load packages, change the page geometry, or redefine the whole heading style.

```
% main.tex
\chapter{Fourier Series}
\input{chapters/fourier-series}

% chapters/fourier-series.tex
\section{Periodic Functions}
Text, equations, figures, and tables for this chapter.
```

Good chapter files begin with the reader's problem, then introduce notation, examples, figures, and exercises. A folder full of small chapter files is easier to revise than one large manuscript file with thousands of lines.

Thesis Decisions

A thesis must satisfy institutional rules. That usually affects page size, margins, font, line spacing, title pages, certificate pages, bibliography style, and annexures. Use `fontspec` only when a system font is required; otherwise pdfLaTeX with `lmodern` is simpler.

Use `biblatex` with `biber` when the bibliography needs modern sorting and citation control. Use `pdfpages` when signed certificates, published papers, or official reports must be included as PDF pages.

Typical Thesis Order

Most thesis problems come from putting formal pages in the wrong place. A practical order is:

1. Title page, certificate, declaration, acknowledgement, and abstract.
2. Table of contents, list of figures, and list of tables.
3. Introduction, literature review, method, results, discussion, and conclusion.
4. Appendices, bibliography, and annexures.

Use Roman page numbers for formal frontmatter when the institution requires it. Start Arabic numbering at Chapter 1. Keep signed certificates and scanned approvals as external PDFs, then include them with `pdfpages` if the final submission requires them.

```
\includepdf[pages=-]{signed-certificate.pdf}
```

Defining a Reusable Box

If the book needs repeated teaching notes such as a principle, warning, or source pattern, define the box once in the preamble and reuse it. The content stays in the chapter; the design stays in one place.

```
\usepackage[most]{tcolorbox}

\newcommand{\principle}[1]{%
  \begin{tcolorbox}[
    title=Principle,
    colback=blue!4,
    colframe=blue!55!black,
    boxrule=0.5pt
  ]
  #1
  \end{tcolorbox}
}

\principle{Write one clear rule here.}
```

CHAPTER 3

Question Papers and Articles

Question papers and research articles are shorter than books and theses, but they still need discipline. A question paper must be scanned quickly. A research article must present a compact scholarly argument.

Choose the Form

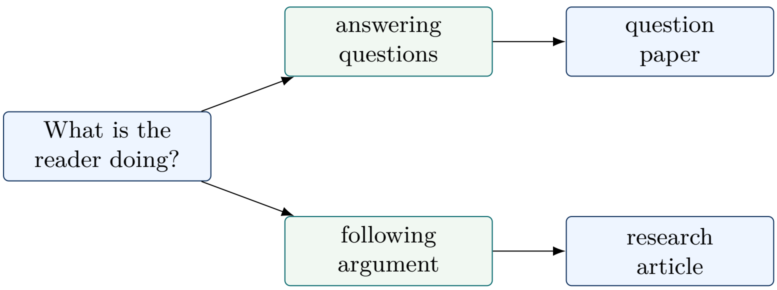


Figure 3.1: The document form follows the reader's task.

Question Papers

The `article` class is usually enough. A question paper needs a header, duration, marks, clear instructions, and questions whose marks are easy to see.

```
\documentclass[12pt]{article}
\usepackage[margin=0.75in]{geometry}
\usepackage{amsmath,array,fancyhdr}
```

Question	Marks
Derive the time period of a simple harmonic oscillator.	5
Explain boundary conditions for a particle in a one-dimensional box.	3
State two assumptions used in the ideal gas model.	2

Use displayed equations when the student must inspect the structure. Keep decoration minimal; an examination page should print cleanly and guide the eye quickly.

Question Paper Anatomy

A question paper is a reading-speed document. Students should find the paper code, time, marks, instructions, and question marks without searching.

Part	What it should contain
Header	Course code, course title, exam name, date, time, and maximum marks.
Instructions	Allowed choices, calculator rules, attempt rules, and assumptions.
Question body	Numbered questions with subparts and visible marks.
Marks column	Right-aligned marks so the examiner and student can scan quickly.
Footer	Page number, paper code, or continuation note when needed.

For physics and mathematics papers, keep variables readable and avoid squeezing long derivations into one line. If the question contains a large expression, display it:

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right), \qquad 0<x<L.$$

```
\[
  \psi_n(x)=\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right),
  \qquad 0<x<L.
\]
```

This is better than inline text because the normalization factor, sine argument, and domain condition can be checked separately.

Use `enumerate` when question numbers matter. Use a table only when you need a stable marks column.

Research Articles

A research article needs title, authors, abstract, sections, equations, figures, tables, and references. It should move quickly from problem to method to result.

```
\documentclass[11pt]{article}
\usepackage{amsmath,amssymb,graphicx,booktabs}
\usepackage[colorlinks=true]{hyperref}
```

Label only equations that are cited later. A label is a promise that the text will return to that item.

```
\begin{equation}\label{eq:energy}
  E_n = \frac{n^2h^2}{8mL^2}.
\end{equation}
Equation~\eqref{eq:energy} shows the quadratic dependence on \(n\).
```

Use BibTeX for simple articles and `biblatex` when citation rules are stricter. In both cases, keep reference data in a `.bib` file.

Article Structure

Most scientific articles can be planned with a simple movement:

Section	Job
Abstract	State the problem, method, main result, and significance in compressed form.
Introduction	Explain the gap, context, and exact question the paper answers.
Method or Theory	Define assumptions, equations, apparatus, data, and procedure.
Results	Present figures, tables, estimates, and comparisons.
Discussion	Interpret the results and name limitations.
Conclusion	State what has been established and what remains open.

The abstract should be written last, even though it appears first. It is a map of the finished argument, not a promise made before the work is clear.

Bibliography Basics

Keep bibliography entries in a separate `.bib` file. The source remains readable, and the same references can be reused in articles, theses, and books.

```
% references.bib
@book{knuth1984texbook,
  author    = {Donald E. Knuth},
  title     = {The TeXbook},
  year      = {1984},
  publisher = {Addison-Wesley}
}

% main.tex
\bibliographystyle{plain}
\bibliography{references}
```

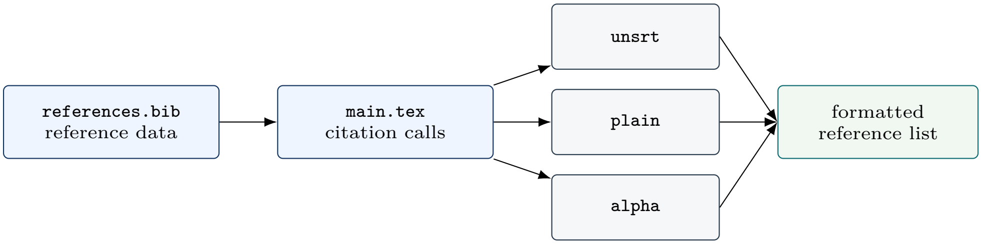


Figure 3.2: The same `.bib` file can produce different bibliography styles by changing the style line in `main.tex`.

Style	Visual effect
`plain`	Sorts references alphabetically and prints numeric labels such as [1].
`unsrt`	Keeps references in citation order and prints numeric labels.
`alpha`	Prints compact author-year-like labels such as [Knu84].
`abbrv`	Similar to `plain`, but abbreviates first names and some fields.

Use `biblatex` when you need stronger control over citation style, sorting, author-year rules, or multilingual bibliographies.

CHAPTER 4

Tables, Figures, and Labels

Tables and figures should explain the document. Labels and references keep those explanations stable when the document changes.

Tables

`tabularx` is useful when one column contains text and must stretch to the line width. `booktabs` gives clean horizontal rules.

```
\begin{tabularx}{\linewidth}{@{}lX@{}}
\toprule
Command & Use \\
\midrule
\input & Insert another source file. \\
\includegraphics & Place an image. \\
\bottomrule
\end{tabularx}
```

Command	Use
`\input`	Insert another source file at the current location.
`\includegraphics`	Place an image, usually inside a figure with a caption and label.
`\label`	Create a reference target.
`\ref`	Print the number of a labelled target.

Equations for Science

Physicists and mathematicians need equations that can be read, referenced, and checked line by line. Use `align` for derivations and related laws; use `cases` for definitions that change by condition.

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \qquad \nabla \cdot \mathbf{B} = 0, \tag{4.1}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \tag{4.2}$$

Equation (4.1) shows the usual pattern: label only the equations the text actually discusses.

```
\begin{align}
  \nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0},
  &
  \nabla \cdot \mathbf{B} &= 0, \label{eq:maxwell-divergence}\\
  \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t},
  &
```

```
  \nabla \times \mathbf{B} &= \mu_0\mathbf{J}
    + \mu_0\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}.
\end{align}
```

Source part	Meaning
`align`	Builds a numbered, aligned display for related equations.
`\&`	Marks alignment points. Here it aligns the equals signs and separates the two columns of laws.
`\textbackslash \textbackslash`	Starts the next numbered row. That is why the display produces equations 4.1 and 4.2.
`\textbackslash label\{eq:maxwell-divergence\}`	Names equation 4.1 so the prose can call it with `\eqref {eq:maxwell-divergence}`.
`\textbackslash notag`	Can be added before a row break when a row should not receive a number.

```
\usepackage{amsthm}
\newtheorem{theorem}{Theorem}[chapter]
\newtheorem{definition}[theorem]{Definition}

\usepackage{siunitx}
\sisetup{per-mode=symbol}
The acceleration is \qty{9.81}{\metre\per\second\squared}.
```

For mathematics-heavy writing, add theorem environments with `amsthm`. For laboratory, physics, and engineering writing, use `siunitx` so numbers and units stay consistent.

Figures

A figure should have a purpose, a caption, and a label. Put the label after the caption so LaTeX attaches the correct number.

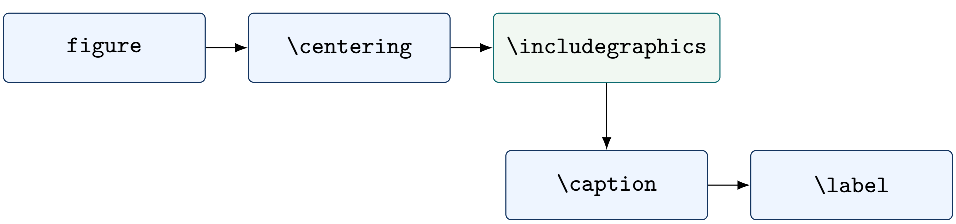


Figure 4.1: A figure float combines image content with a caption and a label.

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.75\linewidth]{experiment}
```

```
  \caption{Experimental arrangement.}
  \label{fig:experiment}
\end{figure}
```

Two or Three Images in One Column

Stack related images when the reader should compare stages, views, or panels from top to bottom. Keep one caption for the whole group unless each panel needs a separate explanation.

```
\begin{figure}[htbp]
  \centering
  \begin{minipage}{0.66\linewidth}
    \centering
    \includegraphics[width=\linewidth]{demo-column-setup-front}

    \vspace{0.35em}
    \includegraphics[width=\linewidth]{demo-column-setup-side}

    \vspace{0.35em}
    \includegraphics[width=\linewidth]{demo-column-data-plot}
  \end{minipage}
  \caption{Three downloaded demo images stacked in one column.}
  \label{fig:three-image-column}
\end{figure}
```

The same source produces a real stacked figure when the image files are present in the graphics path:

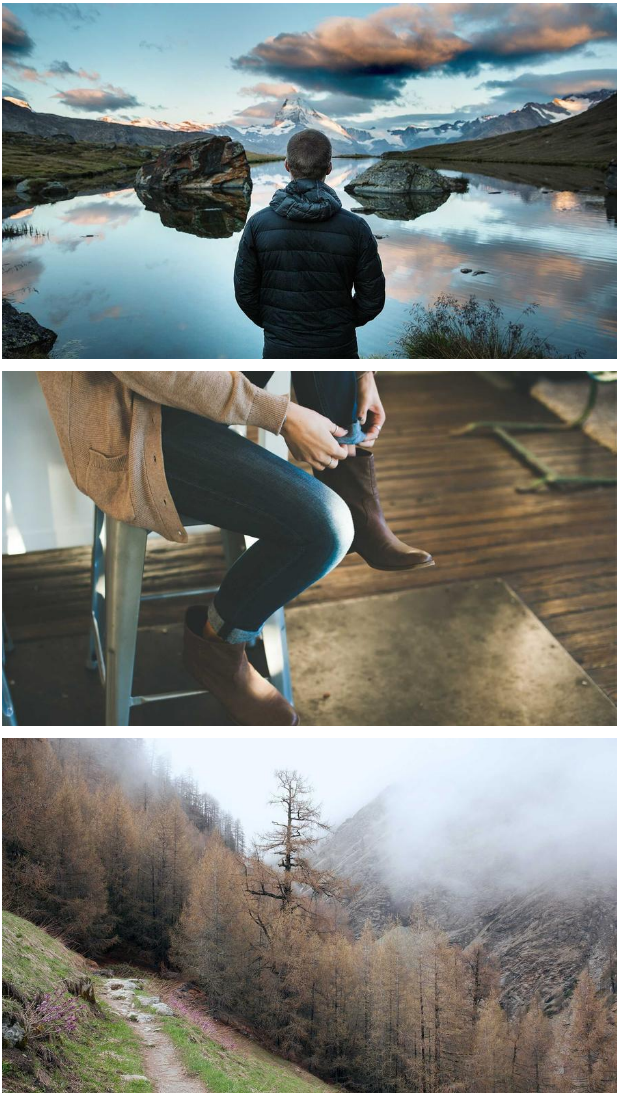


Figure 4.2: Three downloaded demo images stacked in one column.

Use `subcaption` only when the panels need their own labels such as (a), (b), and (c).

Reference Discipline

Labels are part of the document's structure. A label should be unique, meaningful, and placed after the numbered thing it names. Put figure labels after captions, table labels after captions, and equation labels inside the equation environment.

Prefix	Example use
`ch:`	Chapters, such as `\label {ch:methods}`.
`sec:`	Sections, such as `\label {sec:experiment}`.
`eq:`	Equations, such as `\label {eq:schrodinger}`.
`fig:`	Figures, such as `\label {fig:apparatus}`.
`tab:`	Tables, such as `\label {tab:measurements}`.

Use `\eqref` for equations because it prints parentheses in the usual mathematical style. Use `\ref` for chapters, sections, figures, and tables. After adding or moving labels, compile more than once so numbers, contents, and links settle.

Tables for Data

Tables should carry exact values. If the reader needs a trend, use a figure. If the reader needs a number, use a table.

Trial	Voltage	Current	Note
1	1.0	0.12	Stable reading after warm-up.
2	2.0	0.24	Linear response continues.
3	3.0	0.37	Slight deviation at higher setting.

Align numerical columns to the right. Keep long comments in a text column. Do not use vertical rules unless the table is genuinely hard to read without them.

CHAPTER 5

Build, Reuse, and Repair

Compilation is part of writing. The build should be repeatable, generated files should stay separate, and errors should be fixed from the first real cause.

Build Pipeline

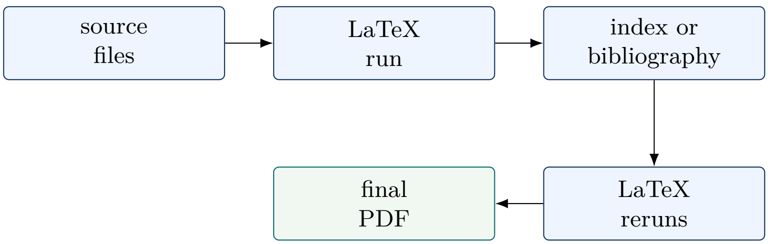


Figure 5.1: Multiple LaTeX runs are normal because references, contents, and indexes settle in stages.

Use the startup commands or `run.sh` to keep auxiliary files in `gar/`. This project's script also copies `gar/main.pdf` to `main.pdf`, so the final PDF appears in the project directory while build files stay separate.

What the Generated Files Mean

Generated files are not junk. They are LaTeX's memory between runs. Keeping them in `gar/` makes the project clean while preserving the information LaTeX needs.

File	Meaning
`.aux`	Cross-reference data, citation requests, labels, and write-back information.
`.toc`	Table-of-contents entries collected from chapters and sections.
`.idx`	Raw index entries written by `\index` commands.
`.ind`	Formatted index created by `makeindex`.
`.log`	The most important diagnostic file; read it when a build fails.
`.out`	Bookmark and outline information for linked PDFs.

When references look wrong, rebuild. When the same error repeats, read the log from the first error line, not from the last warning.

Reuse Without Repeating Mistakes

Reuse stable decisions: page geometry, package loading, heading style, table style, build script, and folder structure. Do not reuse document-specific prose, unfinished formal pages, or old workaround code whose purpose is unclear.

If a design grows, move reusable definitions into a style file such as `bookstyle.sty`. Keep chapter text out of style files.

A Small Style File

When the preamble becomes long, move stable design decisions into a style file. This keeps `main.tex` readable while still making the design reusable.

```
% bookstyle.sty
\ProvidesPackage{bookstyle}
\RequirePackage{xcolor}
\RequirePackage{booktabs}
\RequirePackage{tabularx}

\definecolor{chapterblue}{HTML}{12335F}
\newcommand{\sourcefile}[1]{\texttt{\detokenize{#1}}}
```

Then load it from the main file:

```
\usepackage{bookstyle}
```

Only stable formatting belongs in a style file. Do not hide chapter prose, theorem statements, or project-specific content there.

Repair Table

Message	First place to look
`Undefined control sequence`	Misspelled command or missing package.
`File not found`	Wrong path, wrong extension, or compiling from the wrong folder.
`Missing \$ inserted`	Math syntax used in text mode or an unmatched dollar sign.
`Label(s) may have changed`	Recompile so references and page numbers can settle.
`Overfull hbox`	Long URLs, code, tables, or unbreakable words.

Read the log from the first error. Later errors may only be consequences.

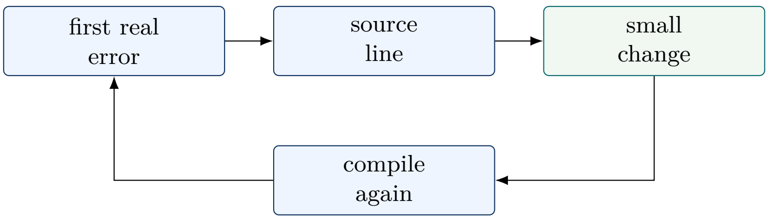


Figure 5.2: Fix one cause at a time; repeated rebuilding is part of the editing loop.

CHAPTER 6

Style Design

Source Pattern

What this chapter solves. This chapter explains how a LaTeX document gets its visual identity: colors, fonts, boxes, headings, headers, tables, code blocks, and presentation themes. The aim is not merely to copy attractive code, but to understand where style decisions belong and why they should be separated from chapter prose.

Working source pattern. `main.tex` loads packages and project-wide definitions; a reusable project may move those definitions into `bookstyle.sty`.

A style is a reusable design decision. In ordinary writing, the author decides what to say. In style design, the author decides how repeated parts of the document should look. A chapter title, a warning box, a source-code block, a table heading, and a running header are not separate problems. They are repeated visual patterns.

The first principle is simple:

Principle

A LaTeX style is not decoration added at the end. It is a system of repeated decisions kept in one place so that the document remains readable, consistent, and easy to revise.

A good style has three jobs. First, it helps the reader recognize the structure of the document. Second, it saves the author from repeating the same formatting commands in every chapter. Third, it makes redesign possible by changing a few definitions instead of editing hundreds of pages.

Start from the Source File

Every LaTeX document has two broad regions: the preamble and the document body.

Region	Purpose
Preamble	The part before `\begin{document}`. It loads packages, defines colors, creates commands, and sets the document style.
Document body	The part between `\begin{document}` and `\end{document}`. It contains chapters, sections, paragraphs, tables, figures, equations, and exercises.

The body should mainly contain content. The preamble should contain the repeated rules that control appearance. When a chapter file contains too many color commands, spacing commands, and box definitions, the author loses sight of the actual subject.

Source Pattern

In this book, `main.tex` currently contains the project-wide definitions: colors such as `chapterblue`, boxes such as `principlebox`, and heading rules defined by `titlesec`. A larger project can move those definitions into a separate style file.

Understand Basic LaTeX Syntax

Style design uses ordinary LaTeX syntax. A style file is not a different language. It stores the same commands and environments that would otherwise be repeated in the preamble.

Syntax	Meaning
`\command`	A command starts with a backslash and asks LaTeX to perform an action.
`{required}`	Braces contain a required argument. The command needs this input to work.
`[optional]`	Square brackets contain optional settings. If omitted, LaTeX uses default values.
`key=value`	Many modern packages use readable option pairs, for example `colback=boxblue`.
`\begin{name}...\end{name}`	An environment applies a structure to a block of content.
`%`	A percent sign starts a comment. LaTeX ignores the rest of that source line.

For example, the command

```
\includegraphics[width=0.75\linewidth]{experiment}
```

has three parts.

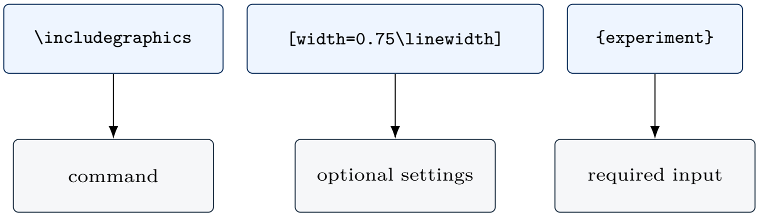


Figure 6.1: A LaTeX command is easiest to understand when its command name, optional settings, and required input are separated.

Source Pattern

The command means: use the image command, set the image width to 75 percent of the current line width, and load the file named `experiment`.

This same logic applies to style commands. A box definition such as `colback=boxblue` is not mysterious. It simply says: use the color named `boxblue` as the background color.

From Repetition to Style

Style design begins when the same formatting pattern appears again and again.

Suppose a manuscript repeatedly uses this kind of note:

```
\begin{tcolorbox}[
  title=Principle,
  colback=boxblue,
  colframe=chapterblue,
  boxrule=0.55pt]
Keep global design decisions in one place.
\end{tcolorbox}
```

This works, but it creates a maintenance problem. If the author later wants a different background color, frame color, title style, or spacing, every box must be edited manually.

The better solution is to name the repeated pattern once:

```
\newtcolorbox{principlebox}[1][]{
  title=Principle,
  colback=boxblue,
  colframe=chapterblue,
  boxrule=0.55pt,
  #1
}
```

Then the chapter file becomes clean:

```
\begin{principlebox}
Keep global design decisions in one place.
\end{principlebox}
```

Principle

A style command is useful when it hides repeated formatting but leaves the meaning of the content visible.

This is the central movement of LaTeX style design:

Stage	What happens
Manual formatting	The author writes the same visual options repeatedly.
Named pattern	The repeated formatting is given a meaningful name.
Reusable style	The named pattern is placed in the preamble or in a style file.
Clean chapter source	The chapter file contains content and semantic commands, not repeated design code.

Preamble or Style File?

For a small document, it is acceptable to keep definitions in `main.tex`. For a book, thesis, course pack, question-paper system, or repeated project, it is better to move stable design definitions into a separate file such as `bookstyle.sty`.

Location	Best use
`main.tex` preamble	Good for one document or for early experimentation.
`bookstyle.sty`	Good for reusable project design shared by many chapters or many documents.
Chapter files	Should contain prose, equations, figures, tables, exercises, and semantic commands. Avoid defining global styles here.

A style file is loaded from the preamble:

```
\usepackage{bookstyle}
```

LaTeX then looks for a file named `bookstyle.sty`. If the file is found, LaTeX reads its definitions before the document body begins.

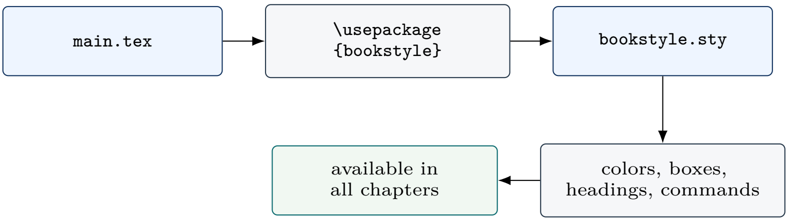


Figure 6.2: A style file is loaded once from the preamble, then its definitions are available throughout the manuscript.

How to Write a Style File

A project style file is a small LaTeX package. It usually has four parts: an identification line, package dependencies, project definitions, and an intentional ending.

```
% bookstyle.sty
\ProvidesPackage{bookstyle}

% 1. Load packages needed by the style.
\RequirePackage{xcolor}
\RequirePackage[most]{tcolorbox}
\RequirePackage{titlesec}
\RequirePackage{fancyhdr}
\RequirePackage{graphicx}
\RequirePackage{tikz}
\RequirePackage{booktabs}
\RequirePackage{tabularx}
\RequirePackage{array}
\RequirePackage{listings}
\RequirePackage{amsthm}
```

```
% 2. Define colors.
\definecolor{chapterblue}{HTML}{12335F}
\definecolor{chapterslate}{HTML}{26384A}
\definecolor{boxblue}{HTML}{EEF5FF}
\definecolor{boxgray}{HTML}{F6F7F9}
\definecolor{accentgold}{HTML}{B77A11}

% 3. Define commands and environments.
\newcommand{\sourcefile}[1]{\texttt{\detokenize{#1}}}

\newtcolorbox{principlebox}[1][]{
  title=Principle,
  colback=boxblue,
  colframe=chapterblue,
  #1
}

% 4. Stop reading here.
\endinput
```

The command used inside a style file is usually `\RequirePackage`, not `\usepackage`. Both load packages, but their roles are different.

Command	Where it belongs
`\usepackage{...}`	Used in a document preamble, normally inside `main.tex`.
`\RequirePackage{...}`	Used inside a package or style file, such as `bookstyle.sty`.

Common Mistake

Do not put chapter paragraphs, exercises, or long explanations inside `bookstyle.sty`. A style file should define appearance. Chapter files should contain content.

The Layered Style System

A style file becomes understandable when it is arranged in layers. The lower layers support the higher layers.

Layer	What belongs there
Package layer	Package loading and library loading. Examples: `xcolor`, `tcolorbox`, `titlesec`, `fancyhdr`.
Token layer	Named colors, lengths, common spacing values, metadata, and file paths.
Command layer	Short reusable commands such as `\sourcefile`, `\term`, or `\BookTitle`.
Environment layer	Reusable boxes and structures such as `principlebox`, `sourcebox`, and `warningbox`.
Page layer	Chapter titles, section headings, running headers, footers, margins, and page spacing.
Subject layer	Special styles for mathematics, theorem environments, bibliography, figures, tables, code listings, or beamer slides.

The order matters. A box style cannot use `chapterblue` before that color has been defined. A theorem style cannot be created before the theorem package has been loaded. A heading style cannot use `titlesec` commands unless `titlesec` has already been loaded.

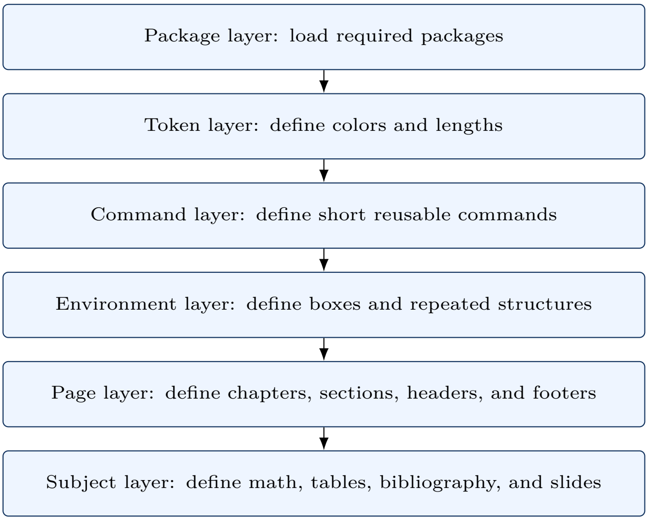


Figure 6.3: A maintainable style file is built from general foundations to visible document structures.

A Minimal Working Style

The smallest useful style system contains one color, one command, one box, and one heading decision. The example below is intentionally small so that the logic is visible.

```
% bookstyle.sty
\ProvidesPackage{bookstyle}
\RequirePackage{xcolor}
\RequirePackage[most]{tcolorbox}
\RequirePackage{titlesec}

\definecolor{chapterblue}{HTML}{12335F}
\definecolor{boxblue}{HTML}{EEF5FF}

\newcommand{\sourcefile}[1]{\texttt{\detokenize{#1}}}

\newtcolorbox{principlebox}[1][]{
  title=Principle,
  colback=boxblue,
  colframe=chapterblue,
  fonttitle=\bfseries,
  #1
}

\titleformat{\section}
  {\normalfont\large\bfseries\color{chapterblue}}
  {}
  {0pt}
  {}
```

```
\endinput
```

After loading this style, the chapter source can remain clean:

```
\section{A Clean Section}

\begin{principlebox}
The chapter file should express meaning. The style file should control repeated appearance.
\end{principlebox}

The file \sourcefile{main.tex} controls the manuscript.
```

Output Preview

A Clean Section

Principle

The chapter file should express meaning. The style file should control repeated appearance.

The file `main.tex` controls the manuscript.

Color Palette

Named colors make a book easier to redesign. Change the value in one place, and every chapter heading, box, rule, and code block follows the new palette.

Swatch	Name	Best use
	`chapterblue`	Chapter titles, strong rules, important labels.
	`accentgold`	Chapter numbers, side rules, small accents.
	`chapterslate`	Secondary headings, table frames, quiet text accents.
	`accentteal`	Reusable-template boxes and process diagrams.
	`boxblue`	Principle and definition box backgrounds.
	`boxgreen`	Practice or reusable-template box backgrounds.
	`boxgold`	Highlights that should feel warm but not loud.
	`boxred`	Warning or common-mistake box backgrounds.
	`boxgray`	Source-pattern panels and neutral chapter-intro panels.
	`codeback`	Listing background.
	`codeframe`	Listing border.

Font Styles

Use pdfLaTeX with `lmodern` for a simple English manuscript. Use XeLaTeX or LuaLaTeX with `fontspec` when the document needs system fonts or Hindi.

Style	Output feel
`lmodern`	A clean TeX-native serif for English books and notes.
`fontspec + Times New Roman`	A familiar manuscript-like serif when system fonts are available.
`TeX Gyre Termes fallback`	A Times-compatible fallback for XeLaTeX or LuaLaTeX.
`monospace`	`Source code, file names, and commands stay visually distinct.`

```
% main.tex, when compiling with pdfLaTeX
\usepackage[T1]{fontenc}
\usepackage[utf8]{inputenc}
\usepackage{lmodern}
```

```
% main.tex, when compiling with XeLaTeX or LuaLaTeX
\usepackage{fontspec}
\IfFontExistsTF{Times New Roman}
  {\setmainfont{Times New Roman}}
  {\setmainfont{TeX Gyre Termes}}
```

Command Shortcuts

Commands are useful when the same phrase or formatting pattern appears many times. Define the command once, then use the short name in chapters. This keeps the chapter source readable and makes later redesign easy.

```
% Simple commands
\newcommand{\BookTitle}{Practical \LaTeX}
\newcommand{\sourcefile}[1]{\texttt{\detokenize{#1}}}
\newcommand{\term}[1]{\textbf{\emph{#1}}}

% Optional argument: first value is used when the option is omitted.
\NewDocumentCommand{\importantnote}{O{Important} m}{%
  \begin{tcolorbox}[title=#1,colback=blue!4,colframe=blue!60!black]
  #2
  \end{tcolorbox}
}
```

Pattern	Use
`\newcommand{\name}{text}`	Stores repeated text or formatting with no argument.
`\newcommand{\name}[1]{...#1...}`	Defines one required argument.
`\NewDocumentCommand`	Handles optional arguments cleanly in modern LaTeX.
`\detokenize`	Prints file paths and source fragments without treating special characters as commands.

Box Styles

Boxes are useful when the same kind of note appears many times. Define the look once, then reuse the environment. The reader sees a stable visual language:

principle boxes for governing ideas, source boxes for code patterns, warning boxes for mistakes, and practice boxes for exercises.

```
% After tcolorbox and colors are loaded once in bookstyle.sty:
\newtcolorbox{definitionbox}[1][]{title=Definition,
  colback=blue!4,
  colframe=blue!60!black,
  boxrule=0.5pt,
  #1}

\newtcolorbox{examplebox}[1][]{title=Example,
  colback=green!4,
  colframe=green!45!black,
  boxrule=0.5pt,
  #1}

\newtcolorbox{warningbox}[1][]{title=Warning,
  colback=red!4,
  colframe=red!60!black,
  boxrule=0.5pt,
  #1}
```

```
\begin{definitionbox}
A macro is a command that expands to stored instructions.
\end{definitionbox}
```

The blue principle box below is made with `tcolorbox`. The title bar, pale background, rounded corners, and frame are all style choices, while the sentence remains ordinary chapter text.

Output Preview

> Principle
>
> The best LaTeX file is not the cleverest file. It is the file that can be opened six months later, understood quickly, compiled reliably, and reused safely.

```
% Copy after the color palette in bookstyle.sty.
\newtcolorbox{principlebox}[1][]{
  enhanced,
  breakable,
  title=Principle,
  colback=boxblue,
  colframe=chapterblue,
  colbacktitle=chapterblue,
  coltitle=white,
  fonttitle=\bfseries,
  arc=1.3mm,
  boxrule=0.55pt,
  left=7pt,
  right=7pt,
  top=7pt,
  bottom=7pt,
  #1
}
```

```
\begin{principlebox}
The best LaTeX file is not the cleverest file. It is the file that can be
```

```
opened six months later, understood quickly, compiled reliably, and reused safely.
\end{principlebox}
```

Heading and Header Styles

Use `titlesec` for visible heading design and `fancyhdr` for running headers and footers.

Output Preview

Book Decisions 18

CHAPTER 2

Books and Theses

Book Decisions

Running headers help readers know where they are; chapter openings help them see where a new unit begins.

```
% After titlesec and fancyhdr are loaded once in bookstyle.sty:
\titleformat{\chapter}[display]
  {\normalfont\bfseries\color{blue!55!black}}
  {\filleft\large\MakeUppercase{\chaptertitlename}\ \thechapter}
  {0.8ex}
  {\titlerule\vspace{1ex}\filright\Huge}
  [\vspace{0.8ex}\titlerule]

\titleformat{\section}
  {\normalfont\large\bfseries\color{blue!55!black}}
  {}
  {0pt}
  {}

\pagestyle{fancy}
\fancyhf{}
\fancyhead[LE,RO]{\thepage}
\fancyhead[LO]{\nouppercase{\rightmark}}
\fancyhead[RE]{\nouppercase{\leftmark}}
\renewcommand{\headrulewidth}{0.4pt}
```

Chapter Opening Style

A chapter opening can combine two parts: `titlesec` controls the chapter number and title, and a small custom command places the summary box under the title. Put the style code in `bookstyle.sty`; use the short command inside each chapter.

Output Preview

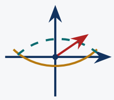
CHAPTER 8

Waves, Optics, and Acoustics Figures

CHAPTER 8

This chapter develops diagram patterns for waves, normal modes, interference, diffraction, ray optics, and acoustics.

Keywords: Wave, Phase, Standing wave, Interference, Diffraction, Lens, Acoustics



```
% After titlesec, tcolorbox, tikz, and colors are loaded once:
\titleformat{\chapter}[display]
  {\normalfont\bfseries\color{chapterblue}}
  {\filleft\large\color{accentgold}\MakeUppercase{\chaptertitlename}\ \thechapter}
  {0.8ex}
  {\titlerule[0.7pt]\vspace{1.0ex}\filright\Huge}
  [\vspace{0.8ex}\titlerule]
\titlespacing*{\chapter}{0pt}{0.5ex}{1.4ex}
```

```
\NewDocumentCommand{\chapterintro}{m m}{%
  \begin{tcolorbox}[
    enhanced,
    colback=boxgray,
    colframe=chapterblue,
    boxrule=0.5pt,
    borderline west={3pt}{0pt}{accentgold},
    arc=1.1mm,
    left=10pt,right=10pt,top=9pt,bottom=9pt]
  {\large\bfseries\color{accentgold}\MakeUppercase{\chaptername}\ \thechapter\par}
  \smallskip
  #1

  \smallskip
  \textbf{\color{chapterblue}Keywords:} #2
  \end{tcolorbox}
}
```

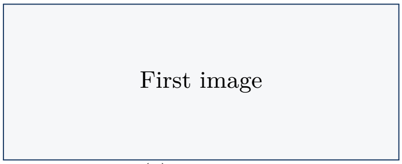
```
% Inside a chapter file
\chapter{Waves, Optics, and Acoustics Figures}
\chapterintro
  {This chapter develops diagram patterns for waves, normal modes,
   interference, diffraction, ray optics, and acoustics.}
  {Wave, Phase, Standing wave, Interference, Diffraction, Lens, Acoustics}
```

Table and Figure Styles

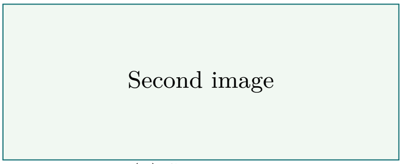
Use `booktabs` and `tabularx` for tables. For images in a row, two panels are safer than three in a narrow book.

Output Preview

Command	Use
`\input`	Insert another source file.
`\includegraphics`	Place an image.



(a) First image



(b) Second image

```
% After booktabs, tabularx, array, and graphicx are loaded once:
\newcolumntype{Y}{>{\raggedright\arraybackslash}X}
\newcommand{\tablehead}[1]{\textbf{#1}}
```

```
\begin{tabularx}{\linewidth}{@{}lY@{}}
\toprule
\tablehead{Command} & \tablehead{Use} \\
\midrule
\verb|\input| & Insert another source file. \\
\verb|\includegraphics| & Place an image. \\
\bottomrule
\end{tabularx}
```

```
\begin{figure}[htbp]
  \centering
  \begin{minipage}{0.48\linewidth}
    \includegraphics[width=\linewidth]{sample-a}
    \caption*{(a) First image}
  \end{minipage}\hfill
  \begin{minipage}{0.48\linewidth}
    \includegraphics[width=\linewidth]{sample-b}
    \caption*{(b) Second image}
  \end{minipage}
  \caption{Two related images shown in one row.}
  \label{fig:two-images-row}
\end{figure}
```

Math and Theorem Styles

For mathematics and physics writing, keep theorem-like structures consistent.

Output Preview

Theorem D.1. *If a wave function is normalized on an interval, its probability density integrates to one.*

$$\int_0^L |\psi_n(x)|^2\,dx = 1, \qquad \psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right).$$

Definition D.2. An eigenvalue is the number produced when an operator acts on an eigenfunction without changing its direction.

Remark D.3. Use theorem styles to make the logical role of each statement visible.

```
% After amsmath, amssymb, mathtools, and amsthm are loaded once:
\theoremstyle{plain}
\newtheorem{theorem}{Theorem}[chapter]
\newtheorem{lemma}[theorem]{Lemma}

\theoremstyle{definition}
\newtheorem{definition}[theorem]{Definition}
\newtheorem{example}[theorem]{Example}

\theoremstyle{remark}
\newtheorem{remark}[theorem]{Remark}
```

Bibliography Styles

For a simple BibTeX document, change only the bibliography style line.

Output Preview

Style	Preview
`plain`	[1] Donald E. Knuth. *The TeXbook*. Addison-Wesley, 1984. Alphabetical by author.
`unsrt`	[1] First cited source. [2] Second cited source. Keeps citation order.
`alpha`	[Knu84] Donald E. Knuth. Compact author-year-like labels.
`biblatex numeric`	[1] Author, title, year, DOI. Better control over sorting, names, DOI, URL, and backend.

```
\bibliographystyle{plain}  % alphabetical numeric list
\bibliography{references}
```

```
\bibliographystyle{unsrt}  % citation-order numeric list
\bibliography{references}
```

```
\bibliographystyle{alpha}  % compact author-year-like labels
\bibliography{references}
```

For stronger control, use `biblatex`:

```
\usepackage[
```

```
  backend=biber,
  style=numeric,
  sorting=none,
  maxnames=3,
  doi=true,
  url=false
]{biblatex}
\addbibresource{references.bib}
```

Beamer Style

For presentations, put theme choices and footer rules in one block near the top of the source.

Output Preview


```
\documentclass[aspectratio=169]{beamer}
\usetheme{Boadilla}
\usecolortheme{seahorse}
\usefonttheme{professionalfonts}
\usepackage{graphicx,amsmath,amssymb,booktabs,hyperref}

\setbeamertemplate{navigation symbols}{}
\setbeamertemplate{blocks}[rounded][shadow=true]

\setbeamertemplate{footline}{
  \leavevmode
  \hbox{
    \begin{beamercolorbox}[wd=.33\paperwidth,ht=2.5ex,dp=1ex,left]{author in head/foot}
      \hspace{1em}\insertsection
    \end{beamercolorbox}
    \begin{beamercolorbox}[wd=.34\paperwidth,ht=2.5ex,dp=1ex,center]{title in head/foot}
      \insertsubsection
    \end{beamercolorbox}
    \begin{beamercolorbox}[wd=.33\paperwidth,ht=2.5ex,dp=1ex,right]{date in head/foot}
      \insertframenumber{} / \inserttotalframenumber\hspace{1em}
    \end{beamercolorbox}
  }
}
```

APPENDIX A

Template Skeletons

These skeletons are starting points. Copy the structure, then replace the names, files, and packages with the needs of the actual document.

Book or Long Manual

```
\documentclass[10pt,openany]{book}
\usepackage[paperwidth=6in,paperheight=9in,
  inner=0.65in,outer=0.50in,top=0.58in,bottom=0.68in]{geometry}
\usepackage{lmodern,amsmath,amssymb,graphicx,booktabs,tabularx}
\usepackage[colorlinks=true]{hyperref}
\usepackage{makeidx}
\makeindex

\begin{document}
\frontmatter
\input{frontmatter/title}
\input{frontmatter/preface}
\tableofcontents

\mainmatter
\chapter{Introduction}
\input{chapters/introduction}

\appendix
\input{appendices/package-reference}

\backmatter
\printindex
\end{document}
```

Use this structure when the document has chapters, frontmatter, appendices, and an index. Keep chapter files focused on writing, not preamble design.

Thesis or Report

```
\documentclass[12pt]{report}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{setspace}
\usepackage{amsmath,amssymb,graphicx,booktabs}
\usepackage[colorlinks=true]{hyperref}
\usepackage[backend=biber,style=numeric]{biblatex}
\addbibresource{references.bib}

\begin{document}
\pagenumbering{roman}
\input{frontmatter/title-page}
\input{frontmatter/declaration}
\input{frontmatter/abstract}
\tableofcontents

\clearpage
\pagenumbering{arabic}
```

```
\chapter{Introduction}
\input{chapters/introduction}

\printbibliography
\end{document}
```

Check institutional rules before changing margins, line spacing, bibliography style, or certificate pages.

Question Paper

```
\documentclass[12pt]{article}
\usepackage[margin=0.75in]{geometry}
\usepackage{amsmath,array,tabularx,fancyhdr}
\pagestyle{fancy}
\fancyhf{}
\lhead{Course Code}
\rhead{Maximum Marks: 50}
\cfoot{\thepage}

\begin{document}
\begin{center}
  \Large Course Title\\
  \normalsize Time: 2 Hours \qquad Maximum Marks: 50
\end{center}

\noindent\textbf{Instructions:} Attempt all questions unless stated otherwise.

\begin{tabularx}{\linewidth}{@{}Xr@{}}
\toprule
Question & Marks \\
\midrule
Derive the expression for the time period of a simple pendulum. & 5 \\
State and explain Gauss's law. & 3 \\
\bottomrule
\end{tabularx}
\end{document}
```

Keep the marks visible. Avoid decorative boxes that make the paper harder to scan or photocopy.

Research Article

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb,mathtools}
\usepackage{graphicx,booktabs}
\usepackage[colorlinks=true]{hyperref}

\title{A Short and Precise Title}
\author{Author Name}
\date{}

\begin{document}
\maketitle
\begin{abstract}
State the problem, method, main result, and significance.
\end{abstract}

\section{Introduction}
\section{Method}
\section{Results}
```

```
\section{Discussion}
\section{Conclusion}

\bibliographystyle{plain}
\bibliography{references}
\end{document}
```

Label only the equations, figures, and tables that the text actually discusses.

Hindi Manuscript

```
\documentclass[12pt]{book}
\usepackage[a5paper,inner=18mm,outer=14mm,top=16mm,bottom=18mm]{geometry}
\usepackage{fontspec}
\setmainfont{Kohinoor Devanagari}[Script=Devanagari]
\setsansfont{Kohinoor Devanagari}[Script=Devanagari]
\usepackage{polyglossia}
\setmainlanguage{hindi}

\begin{document}
\frontmatter
\input{frontmatter/title}
\tableofcontents

\mainmatter
\chapter{Hindi Chapter Title}
\input{chapters/chapter-01}
\end{document}
```

Compile this kind of source with XeLaTeX or LuaLaTeX, not pdfLaTeX.

Beamer Presentation

The progress-report presentation source supplied by the user is a useful Beamer model: a widescreen deck, a simple theme, a custom footer, title-page images, content frames, full-image frames, equations, tables, and a reference frame.

```
\documentclass[aspectratio=169]{beamer}
\usetheme{Boadilla}
\usecolortheme{seahorse}
\usefonttheme{professionalfonts}
\usepackage{graphicx,amsmath,amssymb,booktabs,hyperref}

\setbeamertemplate{navigation symbols}{}
\setbeamertemplate{blocks}[rounded][shadow=true]

\title[Short Title]{Full Presentation Title}
\author{Author Name}
\institute{Department\\University}
\date{\today}

\begin{document}
\begin{frame}[plain]
  \titlepage
\end{frame}

\begin{frame}{Outline}
  \tableofcontents[hideallsubsections]
\end{frame}

\section{Objective}
\begin{frame}{Objective}
  \begin{itemize}
```

```
    \item State the research problem.
    \item State the method.
    \item State the main result or progress.
  \end{itemize}
\end{frame}
\end{document}
```

Beamer part	What it does
`aspectratio=169`	Creates a modern widescreen slide size.
`Boadilla` and `seahorse`	Give a clean academic theme and restrained color set.
`\setbeamertemplate{navigation symbols}{}`	Removes small navigation icons from the slide bottom.
`\tableofcontents[hideallsubsections]`	Builds an outline slide from section commands.
`\begin{frame}{Title}`	Starts one slide with the visible slide title.
`[plain]`	Removes normal headers and footers, useful for title or full-image slides.

```
\begin{frame}[plain]
  \begin{center}
    \includegraphics[width=0.8\paperwidth,
      height=0.8\paperheight]{result-image}
  \end{center}
\end{frame}
```

Figure Blocks

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.75\linewidth]{experiment}
  \caption{Experimental arrangement.}
  \label{fig:experiment}
\end{figure}
```

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.72\linewidth]{sample-a}\\[4pt]
  \includegraphics[width=0.72\linewidth]{sample-b}\\[4pt]
  \includegraphics[width=0.72\linewidth]{sample-c}
  \caption{Three related images shown in one column.}
  \label{fig:image-column}
\end{figure}
```

```
\begin{figure}[htbp]
  \centering
  \begin{minipage}{0.48\linewidth}
    \includegraphics[width=\linewidth]{sample-a}
    \caption*{(a) First image}
  \end{minipage}\hfill
  \begin{minipage}{0.48\linewidth}
    \includegraphics[width=\linewidth]{sample-b}
    \caption*{(b) Second image}
  \end{minipage}
```

```
  \caption{Two related images shown in one row.}
  \label{fig:two-images-row}
\end{figure}
```

Bibliography File Structure

A bibliography becomes easier to maintain when the reference database is separate from the chapter files.

```
project/
  main.tex
  references.bib
  chapters/
    introduction.tex
    method.tex
  figures/
    experiment.pdf
  gar/
    main.pdf
```

```
% main.tex with BibTeX
\documentclass{article}
\usepackage[colorlinks=true]{hyperref}
\begin{document}
Einstein's paper is cited as \cite{einstein1905}.
\bibliographystyle{plain}
\bibliography{references}
\end{document}
```

```
@article{einstein1905,
  author  = {Albert Einstein},
  title   = {On the Electrodynamics of Moving Bodies},
  journal = {Annalen der Physik},
  volume  = {17},
  pages   = {891--921},
  year    = {1905},
  doi     = {10.1002/andp.19053221004}
}

@book{arfken2013,
  author    = {George B. Arfken and Hans J. Weber and Frank E. Harris},
  title     = {Mathematical Methods for Physicists},
  publisher = {Academic Press},
  year      = {2013},
  edition   = {7}
}

@online{latexproject,
  author = {{LaTeX Project}},
  title  = {LaTeX Documentation},
  url    = {https://www.latex-project.org/help/documentation/},
  year   = {2026},
  note   = {Accessed when preparing the manuscript}
}
```

Bibliography Customization

```
% main.tex with biblatex
\usepackage[
  backend=biber,
  style=numeric,
```

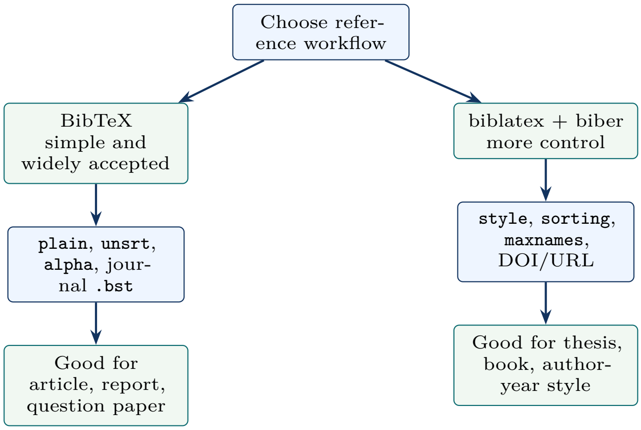


Figure A.1: Bibliography setup has two common routes: BibTeX for simple documents and biblatex with biber for stronger customization.

```
  sorting=none,
  maxnames=3,
  doi=true,
  url=false
]{biblatex}
\addbibresource{references.bib}

\begin{document}
The result is discussed in \cite{einstein1905}.
\printbibliography
\end{document}
```

Option	Effect
`style=numeric`	Prints numbered references such as [1], useful in physics and engineering.
`style=authoryear`	Prints author-year citations, common in humanities and some sciences.
`sorting=none`	Keeps references in the order they are cited.
`sorting=nyt`	Sorts by name, year, and title.
`maxnames=3`	Shows up to three author names before shortening long lists.
`doi=true`	Prints DOI fields when they exist in the `.bib` file.
`url=false`	Hides URL fields for printed books or journal-heavy bibliographies.

APPENDIX B

Package Reference

Package	What it is normally used for
`geometry`	Page size, margins, header space, footer space, and printable text area.
`fontenc`	Font encoding for pdfLaTeX output.
`inputenc`	Input character handling for pdfLaTeX source files.
`lmodern`	Latin Modern fonts for clean pdfLaTeX documents.
`fontspec`	System-font selection with XeLaTeX or LuaLaTeX.
`amsmath`	Alignments, equation environments, and mathematical structure.
`amssymb`	Additional mathematical symbols.
`mathtools`	Improvements and extensions to `amsmath`.
`amsthm`	Theorem, lemma, definition, and proof environments.
`siunitx`	Consistent scientific units, numbers, and measured quantities.
`bm`	Bold mathematical symbols such as vectors and tensors.
`graphicx`	Image insertion, scaling, rotation, and clipping.
`subcaption`	Multiple related images with panel captions.
`xcolor`	Named colors, custom colors, and optional color support inside tables.
`tikz`	Diagrams drawn directly in LaTeX.
`booktabs`	Professional table rules.
`tabularx`	Tables with automatically stretching text columns.
`array`	Better column definitions in tables.
`enumitem`	Control of list spacing and labels.
`listings`	Source-code display with line breaking and syntax highlighting.
`tcolorbox`	Framed teaching boxes, callouts, examples, and designed text panels.
`fancyhdr`	Running headers and footers.
`titlesec`	Chapter and section heading design.
`titletoc`	Table-of-contents design.
`caption`	Caption font, label, and separator control.
`microtype`	Subtle spacing improvements for more polished paragraphs.
`hyperref`	Clickable links, PDF metadata, bookmarks, and reference targets.
`makeidx`	Index creation.
`biblatex`	Modern bibliography management, usually with `biber`.
`pdfpages`	Inclusion of external PDF pages.

APPENDIX C

Math Symbol Tables

These tables collect the commands that are typed most often in physics and mathematics manuscripts. Use them as a writing desk reference rather than as a complete symbol encyclopedia.

Greek Letters

Command	Symbol	Use	Command	Symbol	Use
`\alpha`	α	angle, parameter	`\beta`	β	coefficient
`\gamma`	γ	rate, index	`\Gamma`	Γ	function, width
`\delta`	δ	small change	`\Delta`	Δ	finite change
`\epsilon`	ϵ	small number	`\varepsilon`	ε	permittivity
`\zeta`	ζ	special function	`\eta`	η	efficiency
`\theta`	θ	angle	`\Theta`	Θ	temperature scale
`\vartheta`	ϑ	angle variant	`\iota`	ι	index
`\kappa`	κ	curvature	`\lambda`	λ	wavelength
`\Lambda`	Λ	scale	`\mu`	μ	mean, mass
`\nu`	ν	frequency	`\xi`	ξ	coordinate
`\Xi`	Ξ	particle, sum	`\pi`	π	constant
`\Pi`	Π	product	`\rho`	ρ	density
`\varrho`	ϱ	density variant	`\sigma`	σ	deviation
`\Sigma`	Σ	summation	`\tau`	τ	time constant
`\upsilon`	υ	variable	`\Upsilon`	Υ	particle
`\phi`	ϕ	phase	`\varphi`	φ	phase variant
`\Phi`	Φ	flux	`\chi`	χ	test statistic
`\psi`	ψ	wavefunction	`\Psi`	Ψ	wavefunction
`\omega`	ω	angular frequency	`\Omega`	Ω	ohm, domain

Uppercase Greek letters that look like Latin letters are typed as ordinary Latin letters: A, B, E, Z, H, I, K, M, N, O, P, T, and X.

Common Relations and Operators

Command	Symbol	Meaning	Command	Symbol	Meaning
`=`	$=$	equal	`\neq`	$\neq$	not equal
`\leq`	$\leq$	less or equal	`\geq`	$\geq$	greater or equal
`\ll`	$\ll$	much less	`\gg`	$\gg$	much greater
`\approx`	$\approx$	approximately	`\sim`	$\sim$	similar, scales as
`\equiv`	$\equiv$	identical	`\propto`	$\propto$	proportional
`\pm`	$\pm$	plus or minus	`\mp`	$\mp$	minus or plus
`\cdot`	$\cdot$	dot product	`\times`	$\times$	cross, multiply
`\div`	$\div$	division	`\ast`	$\ast$	convolution marker
`\infty`	∞	infinity	`\partial`	∂	partial derivative
`\nabla`	∇	gradient operator	`\hbar`	$\hbar$	reduced Planck constant

Sets, Calculus, and Logic

Command	Symbol	Use	Command	Symbol	Use
`\in`	$\in$	belongs to	`\notin`	$\notin$	does not belong
`\subset`	$\subset$	subset	`\subseteq`	$\subseteq$	subset or equal
`\cup`	$\cup$	union	`\cap`	$\cap$	intersection
`\emptyset`	$\emptyset$	empty set	`\mathbb{R}`	$\mathbb{R}$	real numbers
`\mathbb{C}`	$\mathbb{C}$	complex numbers	`\forall`	$\forall$	for all
`\exists`	$\exists$	exists	`\Rightarrow`	$\Rightarrow$	implies
`\sum`	$\sum$	summation	`\prod`	$\prod$	product
`\int`	$\int$	integral	`\oint`	$\oint$	closed integral
`\lim`	$\lim$	limit	`\frac{a}{b}`	$\frac{a}{b}$	fraction
`\sqrt{x}`	$\sqrt{x}$	square root	`\sqrt[n]{x}`	$\sqrt[n]{x}$	nth root

Physics Notation

Typed form	Output and use
`\vec{v}`	$\vec{v}$ for a vector with an arrow.
`\mathbf{E}`	$\mathbf{E}$ for a bold vector, common in electromagnetism.
`\hat{x}`	$\hat{x}$ for a unit vector or operator.
`\dot{x}, \ddot{x}`	$\dot{x}$ and $\ddot{x}$ for first and second time derivatives.
`\langle x \rangle`	$\langle x \rangle$ for an expectation value.
`\lvert \psi \rangle`	$\lvert \psi \rangle$ for a quantum ket without an extra package.
`\langle \phi \rvert`	$\langle \phi \rvert$ for a quantum bra without an extra package.
`\Delta x \Delta p \geq \hbar/2`	$\Delta x \, \Delta p \geq \hbar/2$ for the uncertainty relation.

```
\[
  \hat{H}\lvert\psi_n\rangle = E_n\lvert\psi_n\rangle,
  \qquad
  \Delta x\,\Delta p \geq \frac{\hbar}{2}.
\]
```

APPENDIX D

Style Templates

This appendix is a copy-paste reference. The explanations and visual previews are in Chapter 6; the appendix keeps only the compact source blocks that are useful when starting a new book, thesis, article, or presentation.

Principle

Put package loading in one preamble or one `.sty` file. After the packages are loaded, define colors, commands, boxes, headings, tables, figures, math, and bibliography styles without repeating `\RequirePackage`.

Complete Book Style File

Save this file beside `main.tex`. Load it from the main preamble:

```
\usepackage{bookstyle}
```

```
\ProvidesPackage{bookstyle}

\RequirePackage[T1]{fontenc}
\RequirePackage[utf8]{inputenc}
\RequirePackage{lmodern}
\RequirePackage{amsmath,amssymb,mathtools,amsthm}
\RequirePackage[dvipsnames,svgnames]{xcolor}
\RequirePackage{graphicx}
\RequirePackage{tikz}
\RequirePackage{booktabs}
\RequirePackage{array}
\RequirePackage{tabularx}
\RequirePackage{enumitem}
\RequirePackage[most]{tcolorbox}
\RequirePackage{listings}
\RequirePackage{fancyhdr}
\RequirePackage{titlesec}
\RequirePackage{microtype}
\RequirePackage{hyperref}
\RequirePackage{makeidx}

\usetikzlibrary{arrows.meta,calc,decorations.pathreplacing,positioning}
\graphicspath{{assets/images/}}

\definecolor{chapterblue}{HTML}{12335F}
\definecolor{chapterslate}{HTML}{26384A}
\definecolor{boxblue}{HTML}{EEF5FF}
\definecolor{boxgreen}{HTML}{F1F8F2}
\definecolor{boxgold}{HTML}{FFF7E2}
\definecolor{boxred}{HTML}{FFF1EF}
\definecolor{boxgray}{HTML}{F6F7F9}
\definecolor{codeback}{HTML}{F8FAFC}
\definecolor{codeframe}{HTML}{B8C7D9}
\definecolor{accentteal}{HTML}{0B6B70}
\definecolor{accentgold}{HTML}{B77A11}
```

```
\newcommand{\BookTitle}{Practical \LaTeX}
\newcommand{\BookSubtitle}{For Academic and Scientific Documents}
\newcommand{\BookAuthor}{Author Name}
\newcommand{\sourcefile}[1]{\texttt{\detokenize{#1}}}
\newcommand{\term}[1]{\textbf{\emph{#1}}}

\hypersetup{
  colorlinks=true,
  linkcolor=MidnightBlue,
  citecolor=DarkGreen,
  urlcolor=DarkRed,
  pdftitle={Practical LaTeX},
  pdfauthor={Author Name}
}

\lstdefinelanguage{BookLaTeX}{
  language=[LaTeX]TeX,
  morekeywords={frontmatter,mainmatter,backmatter,chapter,section,
    subsection,input,includegraphics,printindex,tableofcontents,
    bibliographystyle,bibliography,addbibresource,printbibliography,
    newcommand,NewDocumentCommand,newtcolorbox,newtheorem,
    titleformat,titlespacing}
}
\lstset{
  language=BookLaTeX,
  basicstyle=\ttfamily\scriptsize,
  keywordstyle=\color{chapterblue}\bfseries,
  commentstyle=\color{SeaGreen!70!black},
  stringstyle=\color{DarkRed},
  backgroundcolor=\color{codeback},
  frame=single,
  rulecolor=\color{codeframe},
  framerule=0.35pt,
  columns=fullflexible,
  keepspaces=true,
  breaklines=true,
  showstringspaces=false
}

\tcbset{
  enhanced,
  breakable,
  arc=1.3mm,
  boxrule=0.55pt,
  left=5pt,
  right=5pt,
  top=5pt,
  bottom=5pt,
  fonttitle=\bfseries
}

\newtcolorbox{principlebox}[1][]{title=Principle,
  colback=boxblue,colframe=chapterblue,
  colbacktitle=chapterblue,coltitle=white,#1}
\newtcolorbox{sourcebox}[1][]{title=Source Pattern,
  colback=boxgray,colframe=chapterslate,#1}
\newtcolorbox{templatebox}[1][]{title=Reusable Template,
  colback=boxgreen,colframe=accentteal,#1}
\newtcolorbox{warningbox}[1][]{title=Common Mistake,
  colback=boxred,colframe=FireBrick,#1}

\titleformat{\chapter}[display]
  {\normalfont\bfseries\color{chapterblue}}
  {\filleft\large\color{accentgold}\MakeUppercase{\chaptertitlename}\ \thechapter}
  {0.8ex}
  {\titlerule[0.7pt]\vspace{1.0ex}\filright\Huge}
  [\vspace{0.8ex}\titlerule]
\titlespacing*{\chapter}{0pt}{0.5ex}{1.4ex}
\titleformat{\section}{\normalfont\large\bfseries\color{chapterblue}}{}{0pt}{}
```

```
\titleformat{\subsection}{\normalfont\normalsize\bfseries}{}{0pt}{}

\pagestyle{fancy}
\fancyhf{}
\fancyhead[LE,RO]{\thepage}
\fancyhead[LO]{\nouppercase{\rightmark}}
\fancyhead[RE]{\nouppercase{\leftmark}}
\renewcommand{\headrulewidth}{0.4pt}

\theoremstyle{plain}
\newtheorem{theorem}{Theorem}[chapter]
\newtheorem{lemma}[theorem]{Lemma}
\theoremstyle{definition}
\newtheorem{definition}[theorem]{Definition}
\newtheorem{example}[theorem]{Example}
\theoremstyle{remark}
\newtheorem{remark}[theorem]{Remark}

\setlist[itemize]{topsep=3pt,itemsep=2pt,leftmargin=1.2em}
\setlist[enumerate]{topsep=3pt,itemsep=3pt,leftmargin=1.4em}
\raggedbottom
\makeindex
\endinput
```

Font Styles

```
% pdfLaTeX
\usepackage[T1]{fontenc}
\usepackage[utf8]{inputenc}
\usepackage{lmodern}
```

```
% XeLaTeX or LuaLaTeX
\usepackage{fontspec}
\IfFontExistsTF{Times New Roman}
  {\setmainfont{Times New Roman}}
  {\setmainfont{TeX Gyre Termes}}
```

Chapter Opening Box

Use this after the colors and `tcolorbox` setup are already loaded.

```
\NewDocumentCommand{\chapterintro}{m m}{%
  \begin{tcolorbox}[
    colback=boxgray,
    colframe=chapterblue,
    boxrule=0.5pt,
    borderline west={3pt}{0pt}{accentgold},
    arc=1.1mm,
    left=10pt,right=10pt,top=9pt,bottom=9pt]
  {\large\bfseries\color{accentgold}\MakeUppercase{\chaptername}\ \thechapter\par}
  \smallskip
  #1

  \smallskip
  \textbf{\color{chapterblue}Keywords:} #2
  \end{tcolorbox}
}
```

```
\chapter{Waves, Optics, and Acoustics Figures}
\chapterintro
  {This chapter develops diagram patterns for waves, optics, and acoustics.}
```

```
  {Wave, Phase, Interference, Diffraction, Lens, Acoustics}
```

Box Styles

```
\newtcolorbox{definitionbox}[1][]{title=Definition,
  colback=boxblue,colframe=chapterblue,#1}
\newtcolorbox{examplebox}[1][]{title=Example,
  colback=boxgreen,colframe=accentteal,#1}
\newtcolorbox{warningbox}[1][]{title=Warning,
  colback=boxred,colframe=FireBrick,#1}
```

```
\begin{definitionbox}
A macro is a command that expands to stored instructions.
\end{definitionbox}
```

Table and Figure Styles

```
\newcolumntype{Y}{>{\raggedright\arraybackslash}X}
\newcommand{\tablehead}[1]{\textbf{#1}}
```

```
\begin{tabularx}{\linewidth}{@{}lY@{}}
\toprule
\tablehead{Command} & \tablehead{Use} \\
\midrule
\verb|\input| & Insert another source file. \\
\verb|\includegraphics| & Place an image. \\
\bottomrule
\end{tabularx}
```

```
\begin{figure}[htbp]
  \centering
  \begin{minipage}{0.48\linewidth}
    \includegraphics[width=\linewidth]{sample-a}
    \caption*{(a) First image}
  \end{minipage}\hfill
  \begin{minipage}{0.48\linewidth}
    \includegraphics[width=\linewidth]{sample-b}
    \caption*{(b) Second image}
  \end{minipage}
  \caption{Two related images shown in one row.}
  \label{fig:two-images-row}
\end{figure}
```

Math and Theorem Styles

```
\theoremstyle{plain}
\newtheorem{theorem}{Theorem}[chapter]
\newtheorem{lemma}[theorem]{Lemma}

\theoremstyle{definition}
\newtheorem{definition}[theorem]{Definition}
\newtheorem{example}[theorem]{Example}

\theoremstyle{remark}
\newtheorem{remark}[theorem]{Remark}
```

```
\[
  \psi_n(x)=\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right),
  \qquad 0<x<L.
\]
```

Bibliography Styles

```
% references.bib
@book{knuth1984texbook,
  author    = {Donald E. Knuth},
  title     = {The TeXbook},
  year      = {1984},
  publisher = {Addison-Wesley}
}
```

```
% main.tex, BibTeX
\bibliographystyle{plain}  % alphabetical numeric list
\bibliography{references}

\bibliographystyle{unsrt}  % citation-order numeric list
\bibliography{references}

\bibliographystyle{alpha}  % compact author-year-like labels
\bibliography{references}
```

```
% main.tex, biblatex
\usepackage[
  backend=biber,
  style=numeric,
  sorting=none,
  maxnames=3,
  doi=true,
  url=false
]{biblatex}
\addbibresource{references.bib}
```

Beamer Style

```
\documentclass[aspectratio=169]{beamer}
\usetheme{Boadilla}
\usecolortheme{seahorse}
\usefonttheme{professionalfonts}
\usepackage{graphicx,amsmath,amssymb,booktabs,hyperref}

\setbeamertemplate{navigation symbols}{}
\setbeamertemplate{blocks}[rounded][shadow=true]
\setbeamertemplate{footline}{
  \leavevmode
  \hbox{
    \begin{beamercolorbox}[wd=.33\paperwidth,ht=2.5ex,dp=1ex,left]{author in head/foot}
      \hspace{1em}\insertsection
    \end{beamercolorbox}
    \begin{beamercolorbox}[wd=.34\paperwidth,ht=2.5ex,dp=1ex,center]{title in head/foot}
      \insertsubsection
    \end{beamercolorbox}
    \begin{beamercolorbox}[wd=.33\paperwidth,ht=2.5ex,dp=1ex,right]{date in head/foot}
      \insertframenumber{} / \inserttotalframenumber\hspace{1em}
    \end{beamercolorbox}
  }
}
```

Further Reading

Use these references when the short explanations in this book are not enough.

***The LaTeX Companion*, Frank Mittelbach and Michel Goossens.** Best for broad LaTeX reference, package choices, and practical explanations.

***The TeXbook*, Donald E. Knuth.** Best for understanding paragraphs, boxes, glue, and page building.

AMS *User's Guide for the amsmath Package.* Best for professional equation layouts and mathematical display structures.

Thomas F. Sturm, *The tcolorbox Package Manual.* Best for designed boxes, breakable examples, and framed teaching material.

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